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UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING H03																							
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV		COVER SHEET		Scale		NTS		= H03	
				Prepared		YR		MV Switchgear for MNA Serang Plant								MVD-01 20kV MAIN SUBSTATION		Customer		PT Multimas Nabati Asahan		& 306	
00		Issued for Approval		02.01.19		YR		Checked		AW						SCHEMATIC DIAGRAM		Drawing Nbr.		18.012-APS-302.02		Sh. 1	
Ind.		Revision		Date		Name		Approved		AI		Based on: BE318042				PT. ABB Sakti Industri		SBC-W OUTGOING		Customer Doc. Nbr.		Cont. 27	
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UniSec

Medium Voltage Switchgear
PT MULTIMAS NABATI ASAHAN
MVD-01 20kV MAIN SUBSTATION
SCHEMATIC DIAGRAM
SBC-W OUTGOING
H03

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A	SHEET INDEX								A						
	PAGE		PAGE NAME		REVISION										
	==MVD-01=H03&306/1		COVER SHEET		00										
	==MVD-01=H03&306/2		Index of Sheet		00										
	==MVD-01=H03&306/3		REFERENCE DESIGNATIONS		00										
	==MVD-01=H03&306/4		REFERENCE DESIGNATIONS		00										
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	==MVD-01=H03&306/7		DISTRIBUTION OF AUXILIARY CIRCUITS		00										
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==MVD-01=H03&306/9		CIRCUIT BREAKER CONTROL		00											
==MVD-01=H03&306/10		AUXILIARY MOTOR & RELAY		00											
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	==MVD-01=H03&306/18		Terminal diagram =H03+-XDA		00										
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==MVD-01=H03&306/26		Terminal diagram =H03+-XDV		00											
==MVD-01=H03&306/27		Terminal diagram =H03+-XDX		00											
E									E						
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	Index of Sheet MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H03	
					Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 306	
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	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS																				-BGB5 POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE ENERGIZED																			
B	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)																				-BGB6 POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE EXCLUDED MECHANICALLY																			
	DESIGNATION																				DESCRIPTION																			
C	-AA MULTIFUNCTION UNIT (CENTRAL UNIT)																				-BGB7 POSITION SWITCHES OF CIRCUIT-BREAKER ON SWITCHGEAR																			
	-BAD1,-BAD4 CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1																				-BGB8...-BGB10 POSITION SWITCHES OF CIRCUIT-BREAKER																			
	-BAD2,-BAD5 CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2																				-BGB11 POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING																			
	-BAD3,-BAD6 CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3																				-BGB12 PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION																			
	-BAR VOLTAGE PROTECTION RELAY																				-BGB13 PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION																			
	-BAS1 VOLTAGE SENSOR LOCATED ON PHASE L1																				-BGD POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.																			
	-BAS2 VOLTAGE SENSOR LOCATED ON PHASE L2																				-BGD1 POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR																			
	-BAS3 VOLTAGE SENSOR LOCATED ON PHASE L3																				-BGD2 POSITION SWITCH OF CABLE COMPARTMENT DOOR																			
	-BAT1 VOLTAGE TRANSFORMER LOCATED ON PHASE L1																				-BGE1 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION																			
	-BAT2 VOLTAGE TRANSFORMER LOCATED ON PHASE L2																				-BGE2 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION																			
D	-BAT3 VOLTAGE TRANSFORMER LOCATED ON PHASE L3																				-BGE3 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BAT4 VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1																				-BGE4 POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE ACTUATED BY ELECTRIC MOTOR																			
	-BAT5 VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2																				-BGE5 POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 ACTUATED BY ELECTRIC MOTOR																			
	-BAT6 VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3																				-BGE7 POSITION SWITCH SIGNALLING EARTHING SWITCH NOT IN MANUAL OPERATION																			
	-BCD DIFFERENTIAL PROTECTIVE RELAY																				-BGE8 POSITION SWITCHES OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH- DISCONNECTOR -QBS																			
	-BCF FEEDER PROTECTION RELAY																				-BGE11 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION																			
	-BCG GENERATOR PROTECTION RELAY																				-BGE12 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN CLOSED POSITION																			
	-BCM MOTOR PROTECTION RELAY																				-BGE13 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BCN NEUTRAL (RESIDUAL) CURRENT TRANSFORMER																				-BGE14, -BGE15 POSITION SWITCHES FOR OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS, ACTUATED BY ELECTRIC MOTOR																			
	-BCP TRANSFORMER PROTECTION RELAY																				-BGF POSITION SWITCHES OF MEDIUM VOLTAGE FUSES																			
E	-BCR CURRENT PROTECTION RELAY																				-BGI1 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION																			
	-BCS1 CURRENT SENSOR LOCATED ON PHASE L1																				-BGI2 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION																			
	-BCS2 CURRENT SENSOR LOCATED ON PHASE L2																				-BGI3 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BCS3 CURRENT SENSOR LOCATED ON PHASE L3																				-BGI4 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR																			
	-BCT1 CURRENT TRANSFORMER LOCATED ON PHASE L1																				-BGI5 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR																			
	-BCT2 CURRENT TRANSFORMER LOCATED ON PHASE L2																				-BGI6 POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION																			
	-BCT3 CURRENT TRANSFORMER LOCATED ON PHASE L3																				-BGI7 POSITION SWITCH SIGNALLING DICONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION																			
	-BCT4,-BCT7 CURRENT TRANSFORMER LOCATED ON PHASE L1																				-BGI8 POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS																			
	-BCT5,-BCT8 CURRENT TRANSFORMER LOCATED ON PHASE L2																				-BGI9 POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR																			
	-BCT6,-BCT9 CURRENT TRANSFORMER LOCATED ON PHASE L3																				-BGI11 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION																			
F	-BCZ DISTANCE PROTECTION RELAY																				-BGI12 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION																			
	-BEF FREQUENCY PROTECTION RELAY																				-BGI13 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BER SUPERVISION RELAYS																				-BGI14, -BGI15 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR																			
	-BES SYNCHRONIZING RELAY																				-BGI21 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION																			
	-BET THERMAL PROTECTION RELAY																				-BGI22 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION																			
	-BGA1 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT																				-BGI23 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BGA2 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)																				-BGI24, -BGI25 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR																			
	-BGA3 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT																																							
	-BGA4 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)																																							
	-BGB1...-BGB3 POSITION SWITCHES OF CIRCUIT-BREAKER																																							
-BGB4 POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS																																								
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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS							A
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT							
B	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							B
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT							
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT							
	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							
C	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							C
	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST							
	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
D	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							D
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							
	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							
E	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							E
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2							
	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3							
F	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							F
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2							
G	-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							G
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							
H	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							H
	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK							
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							
I	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							I
	-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							
	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							
J	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							J
	-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)							
K	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							K
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							
L								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							L
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							
M								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							M
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING		Scale	NTS	= H03	F	
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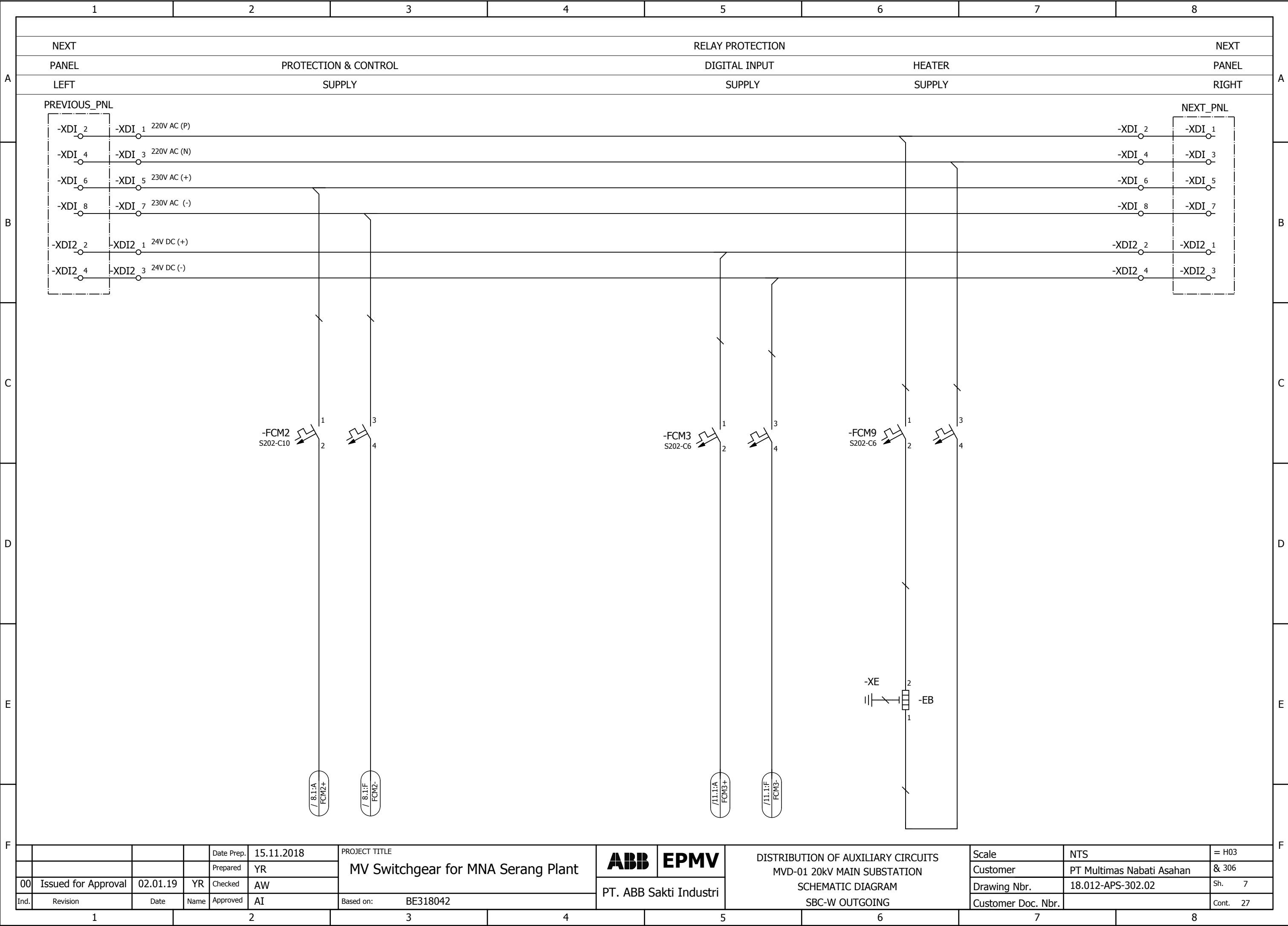
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS						
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS						
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS						
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS						
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK						
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING						
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH						
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)						
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)						
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES						
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR						
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES						
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS						
	-PFF	FLAG RELAYS						-RAR	RESISTORS						
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS						
	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING						
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT						
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE						
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION						
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD						
C	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1						
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2						
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS						
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4						
	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5						
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES						
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS						
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS						
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS						
D	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS						
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES						
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS						
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS						
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES						
	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS						
	-PGV	VOLTMETERS						-TB	CONVERTER						
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE						
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE						
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE						
E	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING						
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT						
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE						
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)													
	-QBD4	TWO-WAY DISCONNECTOR													
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER													
1		2		3		4		5		6		7		8	
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING		Scale	NTS		= H03
				Prepared	YR							Customer	PT Multimas Nabati Asahan		& 306
00	Issued for Approval	02.01.19	YR	Checked	AW							Drawing Nbr.	18.012-APS-302.02		Sh. 5
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		PT. ABB Sakti Industri				Customer Doc. Nbr.		Cont. 27	

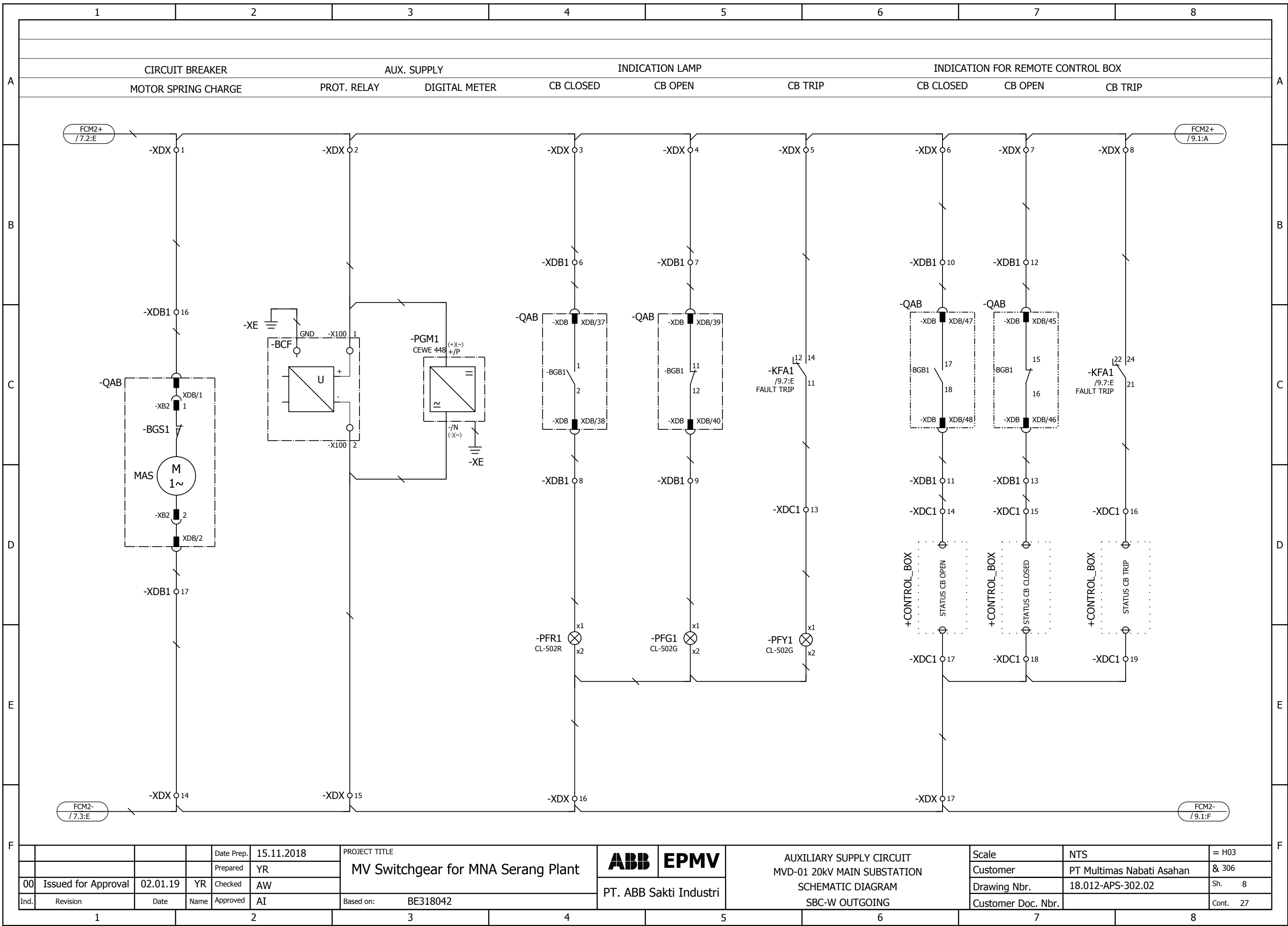
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A	-TFA				ACTIVE POWER TRANSDUCERS				-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS				A	
	-TFC				CURRENT TRANSDUCERS					-XDI2	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING				
	-TFF				FREQUENCY TRANSDUCERS				-XDI3		TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION				
	-TFJ				ACTIVE ENERGY TRANSDUCERS					-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS				
	-TFK				REACTIVE ENERGY TRANSDUCERS				-XDI6		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS				
	-TFM				MULTIFUNCTION TRANSDUCERS					-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS				
	-TFP				POWER-FACTOR TRANSDUCERS				-XDI8		TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS				
	-TFQ				REACTIVE POWER TRANSDUCERS					-XDI9	TERMINAL BLOCK FOR SPECIAL USE				
	-TFS				SIGNAL CONVERTERS				-XDJ		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD				
	-TFV				VOLTAGE TRANSDUCERS					-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS				
B	-WA				BUSBARS				-XDM		SEALABLE TERMINAL BLOCK FOR MEASUREMENT				B
	-WBC				POWER CABLES					-XDS	CURRENT SOCKETS				
	-WE				EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.				-XDT		TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB				
	-WGC				CONTROL CABLES					-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK				
	-WGS				COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)				-XDU1		TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK				
	-WH				OPTICAL FIBRES					-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS				
	-XDA				TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS				-XDV1		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1				
	-XDA1				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1					-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2				
	-XDA2				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2				-XDV3		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3				
	-XDA3				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3					-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR				
	-XDA5				CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER				-XDX		SUPPORT TERMINAL BLOCK				
	-XDB				CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS					-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE				
	-XDB1				CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)				-XE2		EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE				
	-XDB2				CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER					-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE				
	-XDB10...-XDB89				CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS				-XE6		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE				
	-XDB7				CONNECTOR FOR TRUCK POSITION CONTACTS					-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE				
	-XDB91				CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR				-XE8		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE				
	-XDB92				CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR					-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT				
	-XDB93				CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR										
	-XDB95				CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET										
D	-XDC				CUSTOMER TERMINAL BLOCK										
	-XDC1				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER										
	-XDC2				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR										
	-XDC3				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT										
	-XDC4				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS										
	-XDC5				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER										
	-XDE				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE										
	-XDE1				TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE										
	-XDE2				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2										
	-XDE3				TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2										
	-XDF				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS										
	-XDF1				CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR										
	-XDF2				CONNECTOR FOR REAR VENTILATOR CIRCUITS										
	-XDF3				CONNECTOR FOR FRONT VENTILATOR CIRCUITS										
	-XDH				TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)										
	-XDI				TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)										
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB		EPMV	REFERENCE DESIGNATIONS		Scale	NTS
					Prepared	YR	MV Switchgear for MNA Serang Plant			MVD-01 20kV MAIN SUBSTATION		Customer	PT Multimas Nabati Asahan	& 306	
	00	Issued for Approval	02.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM		Drawing Nbr.	18.012-APS-302.02	Sh. 6
	Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				SBC-W OUTGOING		Customer Doc. Nbr.		Cont. 27
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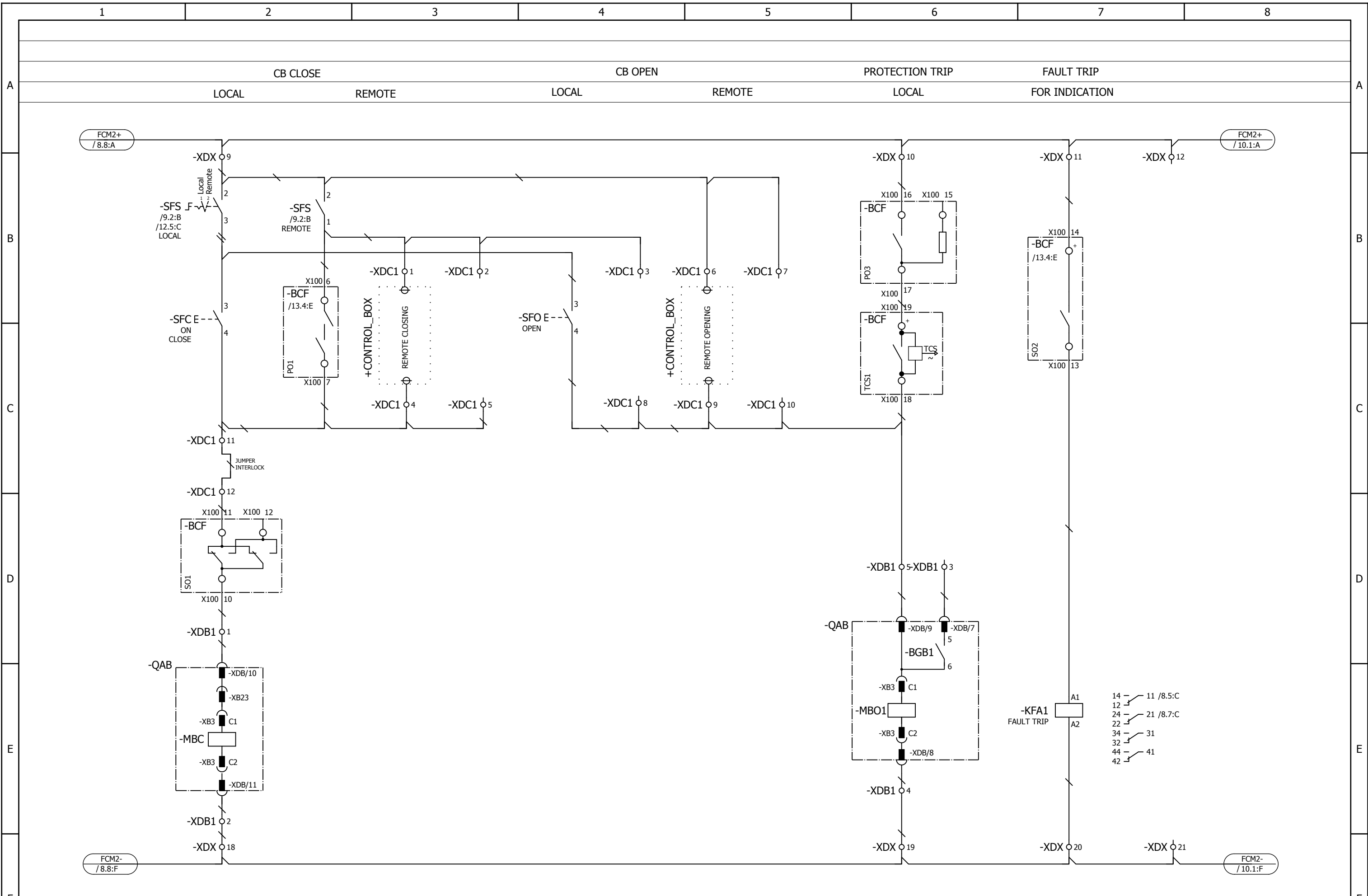
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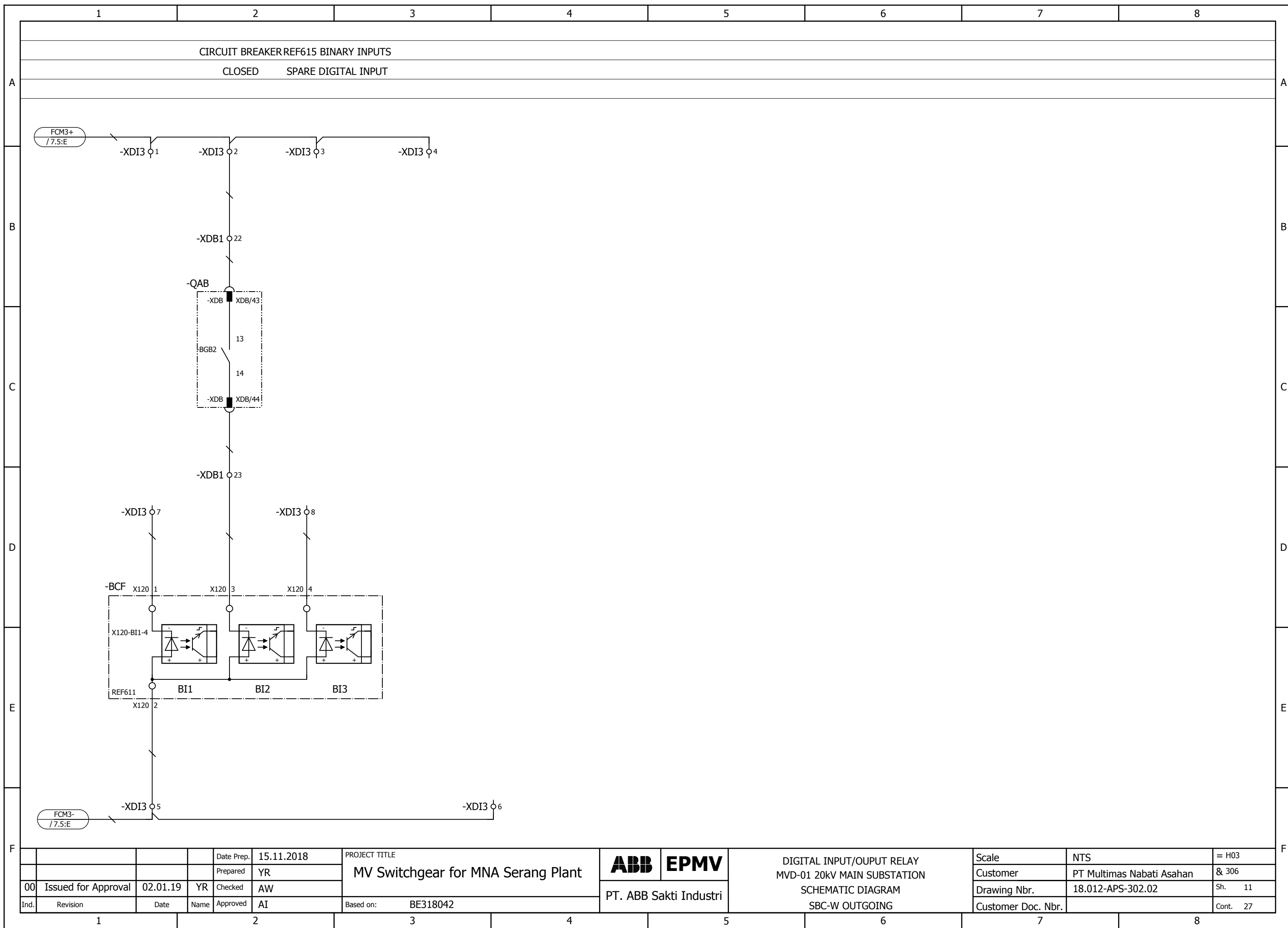


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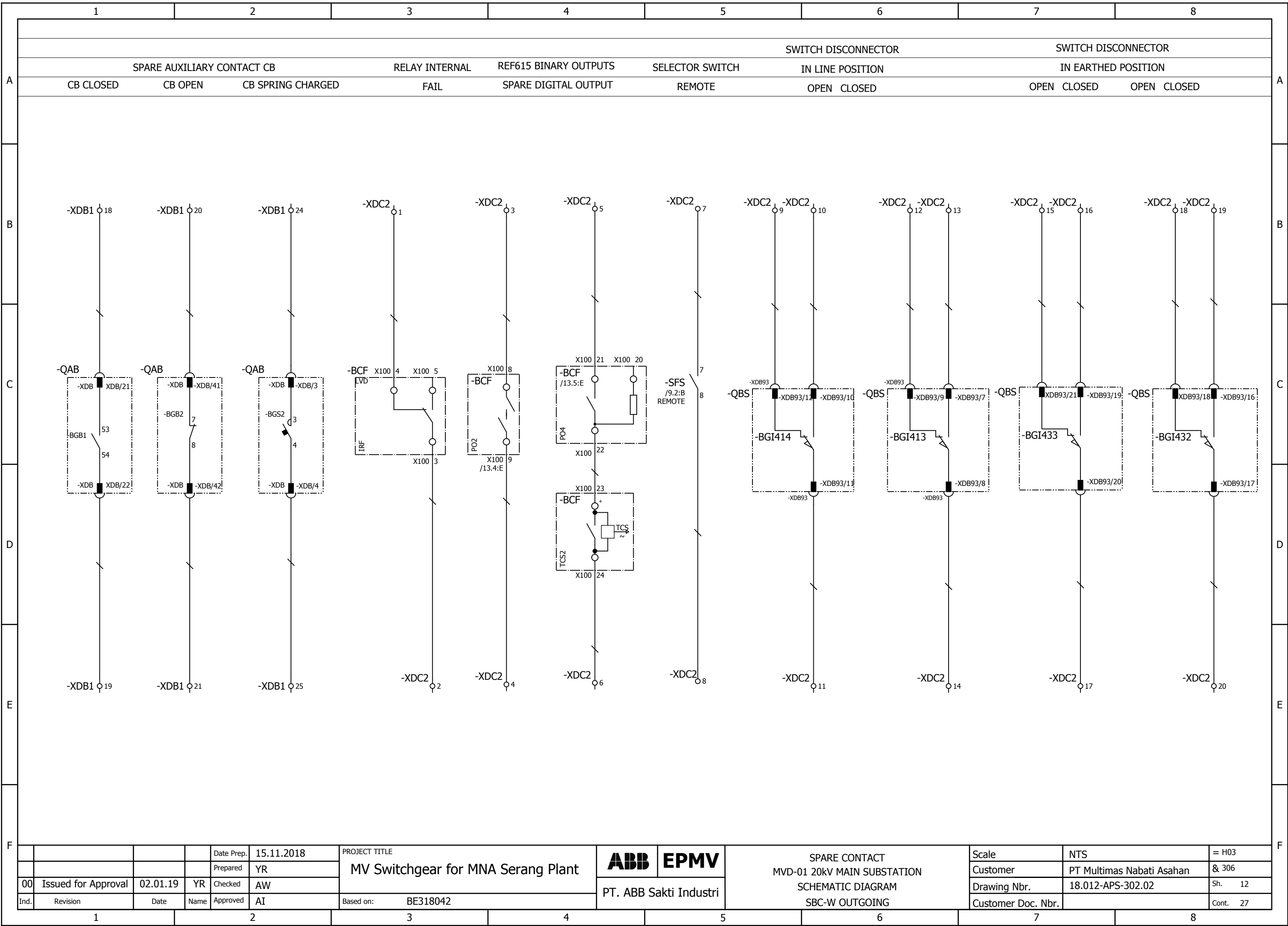


		Date Prep.		15.11.2018	PROJECT TITLE		ABB EPMV		CIRCUIT BREAKER CONTROL MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING		Scale	NTS	= H03
		Prepared		YR							Customer	PT Multimas Nabati Asahan	& 306
00 Issued for Approval		02.01.19	YR	Checked							Drawing Nbr.	18.012-APS-302.02	Sh. 9
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042		PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 27

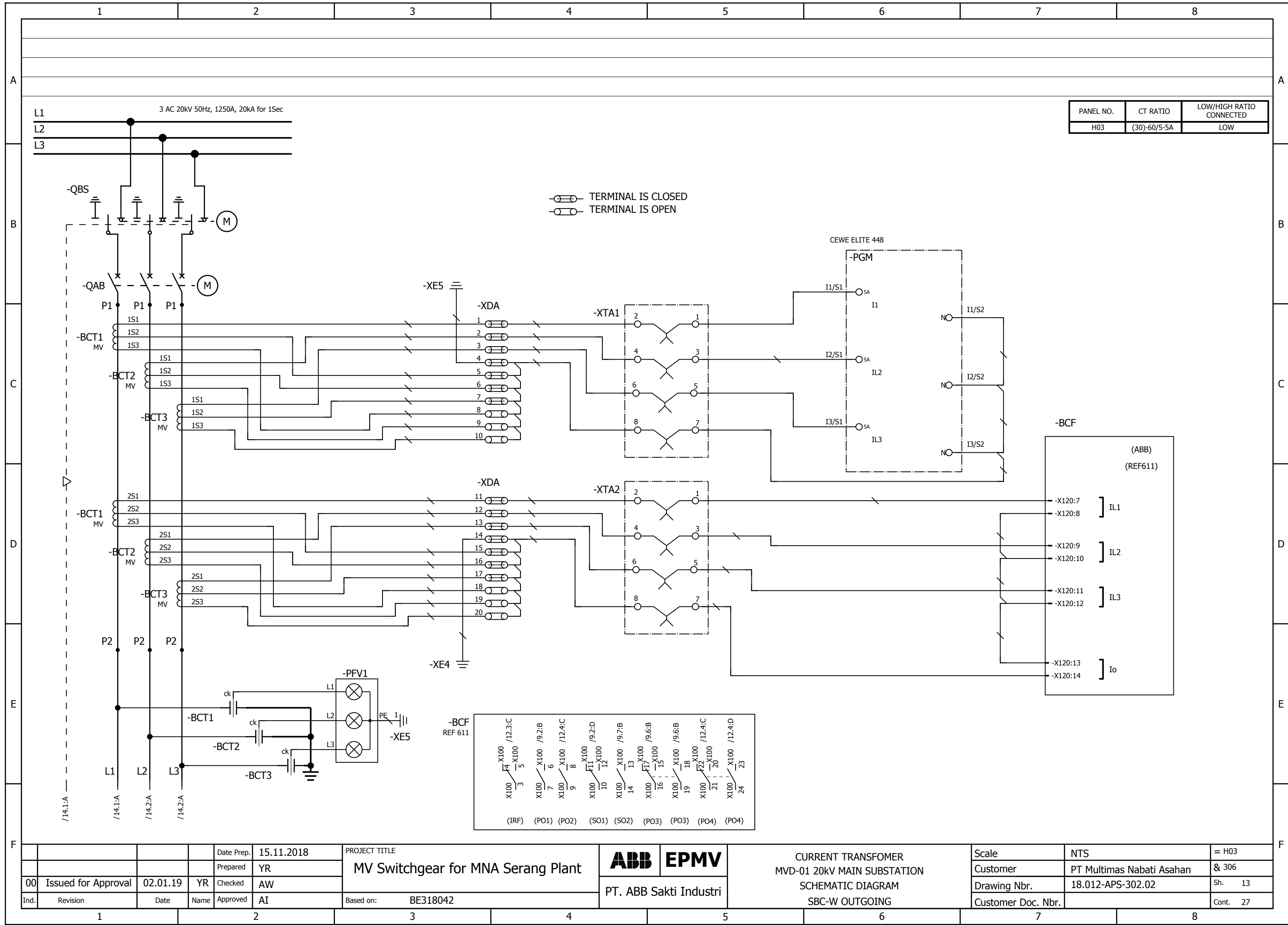
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	AUXILIARY MOTOR & RELAY MVD-01 20KV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H03
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	02.01.19	YR	Checked	AW		Drawing Nbr.	18.012-APS-302.02		Sh. 10		
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.		Cont. 27		
							PT. ABB Sakti Industri					



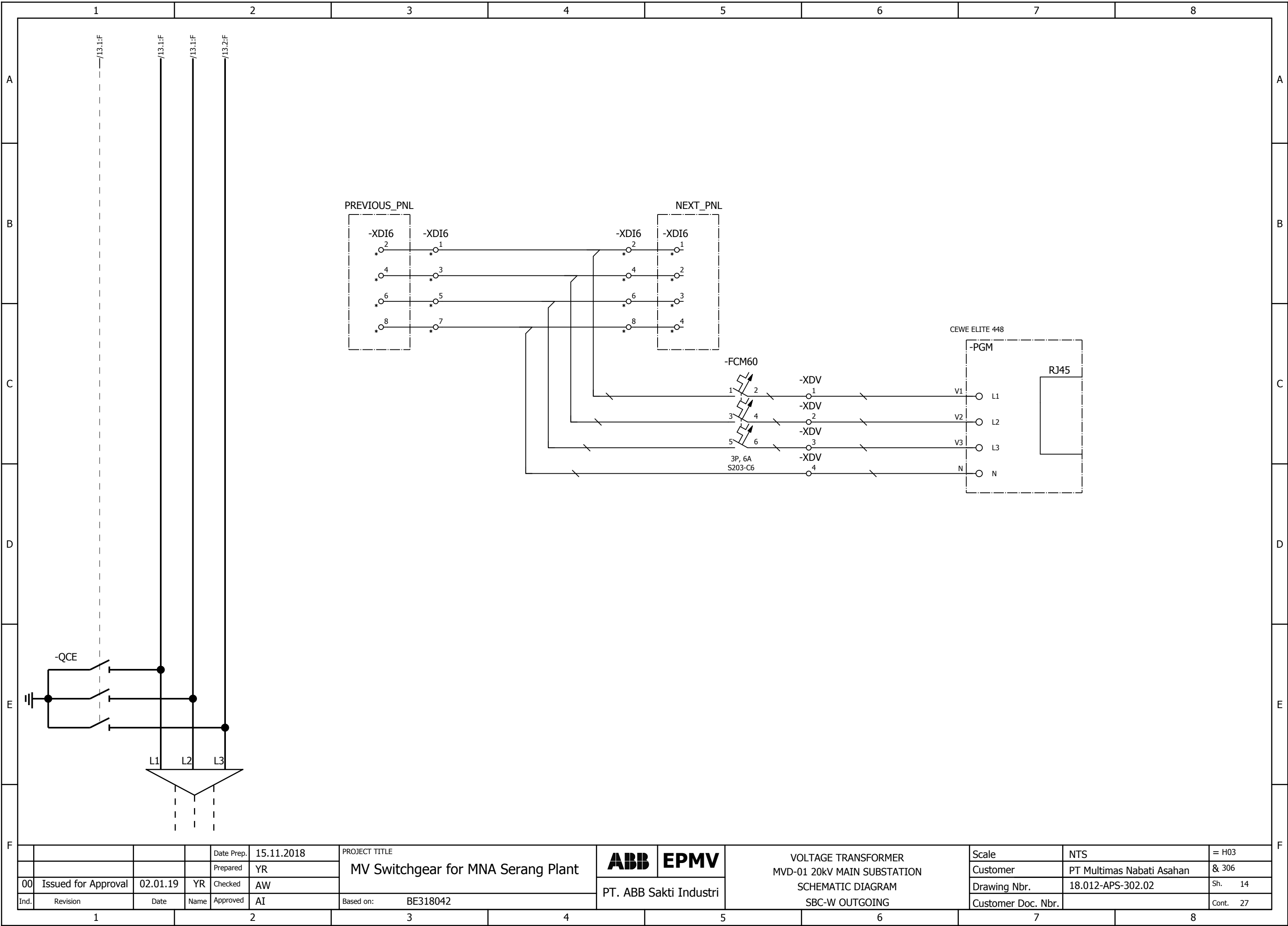
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	VOLTAGE TRANSFORMER MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H03
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	02.01.19	YR	Checked	AW		Drawing Nbr.	18.012-APS-302.02		Sh. 14		
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							PT. ABB Sakti Industri					

No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H03
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	02.01.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		18.012-APS-302.02	Sh.	16
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		Customer Doc. Nbr.			Cont.	27

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A	Summarized parts list							IDABB_F02_005-IDABB							
								Carry over: 0.00							
	TAG ID	DESCRIPTION			MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS							
	-BCF	Feeder Management Relay REF611HBAABC1NN11G			ABB	REF611HBAABC1NN11G	1	/8.2:C							
	-EB	HEATER 230VAC, 50W			SANYANG	CDXN010033P0009	1	/7.6:E							
	-FCM2	MCB S202-C10 MCB. 2P, 10A			ABB	2CDS252001R0104	1	/7.2:C							
	-FCM3;-FCM9	MCB S202-C6 MCB. 2P, 6A			ABB	2CDS252001R0064	2	/7.5:C;/7.6:C							
	-FCM60	MCB S203-C6 MCB. 3P, 6A			ABB	2CDS253001R0064	1	/14.5:C							
	-KFA1	Auxiliary Relay 230AC 4COS			PHOENIX	1YRA3DL266230001	1	/9.7:E							
	-PFG1	PILOT LIGHT GREEN #CL-523G#230VAC CL-523G			ABB	1SFA619402R5232	1	/8.5:E							
B	-PFR1	PILOT LIGHT RED #CL-523R#230VAC CL-523R			ABB	1SFA619402R5231	1	/8.4:E							
	-PFY1	PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y			ABB	1SFA619402R5233	1	/8.6:E							
	-PGM1	POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC			CEWE	1YRA4EA0350001	1	/8.3:C							
	-SFC	PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR			IDEC	1YRA3DA14700001	1	/9.2:B							
	-SFO	PUSH BUTTON, YW1B-M1E10-R RED COLOUR			IDEC	1YRA3DA14700002	1	/9.4:B							
C	-SFS	LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)			K&N	1YRA3DA17940011	1	/9.2:B							
	-XTA1;-XTA2	Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting			YONGSUNG	YSCTT-04C	2	/13.4:C;/13.4:D							
D															
E															
F															
				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	PART LIST		Scale	NTS	= H03	
				Prepared	YR	MV Switchgear for MNA Serang Plant				MVD-01 20kV MAIN SUBSTATION		Customer	PT Multimas Nabati Asahan	& 306	
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